Physical Account (D6) -- Annex C: every number we assumed, narrowed or overrode, and where the code uses it

Provenance tags: OFFICIAL_BRIEF (Annex A/B), ORGANIZER_ERRATA (Graziano clarifications: G4 = 990 kg, Scenario B = G1+G3+G4+G5 = 9,560 kg), ASSUMED (ours, also in ASSUMPTIONS.yaml). Only ASSUMED/narrowed values are listed below.

Resource class	What we assumed	Value / range	Why	Where used in code
Electrical power	Guest reserve at cruise; reserve after one genset trip; mission line lost on trip	200 kW; 100 kW; no feeder lost (trip cuts the shared reserve, both lines stay up)	Brief gives the reserve, not the feeder topology; cutting the shared total is the case D4 must survive	vessel_baseline.yaml:8-9, budgets.py::evaluate_eight_class_budgets, degradation.py::execute_graceful_degradation
Electrical power	Line assignment and per-line limit; peak tolerance	LINE_A_FWD (bay1, mast) / LINE_B_AFT (bay2, hangar, roof, deck, ramp), 150 kW each; peaks counted as committed load, no overdraw tolerance	Splits fwd/aft by position; zero tolerance is fail-safe	vessel_baseline.yaml:10-16, budgets.py::evaluate_eight_class_budgets
Electrical power	Firm reservation vs actual draw	G3 25 kW recharge and G4 35 kW recovery are committed before they draw	A plan the ship cannot power must not be approved	ship_state.py::FirmReservedLoad, replay.py::run_scenario_b_replay
Heat rejection	Seawater temperature; stale feed	A.10: 24 C cruise, 30 C hot survey/degraded; stale or missing feed -> 30 C worst case	Judge against the voyage, never the 16 C pier	mission_envelope_a10.yaml, ship_state.py:259, scheduling.py:684-690
Heat rejection	Out-of-range C(T); loop loss; inertia; kW-to-heat	Clamp 160 kW below 12 C; loop fault x0.60 (72 kW at 30 C); +30 kW x 60 s (1.8 MJ, 600 s recharge), never for sustained loads; 1 kW electric = 1 kW heat	Conservative, energy conserved 1:1	vessel_baseline.yaml:18-33, budgets.py:54 compute_cooling_water_capacity_w
Heat rejection	Air-cooled domains	mast_3 2,000 W (brief); hangar_roof 2,000 W (ASM-ROOF-AIR-2000W); G5 antenna 1,200 W full / 500 W degraded	Brief gives no roof air limit; we copy the mast one	vessel_baseline.yaml:194-198, G5_emergency_comms_edge.yaml
Positions	Bay-sharing rule; mast eligibility; roof vs drone	bay1 splits into bay1_a/bay1_b; bay2 alone 4,000 kg; flight deck and stern ramp exclusive; only mast_3 free; G5 antenna on roof allowed only under ROOF_OUTSIDE_G4_RECOVERY_V1 (conditional, enforced: unverified/invalidated stamp -> D1 YES_IF, D3 refuses every recovery plan)	Graziano: declare the roof/drone conflict or an enforced condition; we enforce	vessel_baseline.yaml:51-80,200-211, budgets.py:97 evaluate_roof_drone_clearance, scheduling.py:187 evaluate_roof_drone_clearance_safe, arbitration.py::_evaluate_candidate_feasibility_and_score, replay.py::_apply_due_d3_milestones
Positions	Stern-ramp override (narrowed brief)	ramp craft 5,000 kg, cradle-assisted 5,500 kg	Brief 5 t; +500 kg only with the cradle, declared as an override	vessel_baseline.yaml:106-107, budgets.py:839
Mass / stability	Multipliers; margin kept	deck/bays x1.0, hangar/roof/deck x1.2, mast x3.0; limits 10,000 kg and 12,000 kg-eq, 0 kg extra margin (Scenario B keeps 440 kg / 2,147 kg-eq)	Brief limits used as hard lines; no hidden buffer	vessel_baseline.yaml:125-130, budgets.py:976-1001
Data links	SATCOM availability; EMCON	8 Mbit/s nominal, 6 cruise, 4 hot survey, 2 degraded, floor 1 Mbit/s (below brief 2 -> stricter), 0 in EMCON	Brief: shared 2-8 Mbit/s, 70-95 % availability; we budget the low end per window	mission_envelope_a10.yaml, vessel_baseline.yaml:136-139, budgets.py:1015
Data links	Acoustic drop model (G3)	HOLD handshake 10-60 s, 15 s nominal; no ACK by 60 s -> plan escalated, ship does not turn	Brief A.5: acoustic 10-60 s latency	vessel_baseline.yaml:185-186, replay.py:1047, G3_auv_survey.yaml
State transitions	Time to go quiet; course/speed; time to clear deck per crew	quiet 180 s; settle into wind 90 s; recovery at 5 kn; deck clear 240 s (14) / 600 s (6) / forbidden (0)	A.6 ranges, mid values chosen	scheduling.py:15-30 DEFAULT_TRANSITION_TIMING_POLICY
Compute / response time	Per-guest allocation; solver budget and timeout	4 nodes x 80 % allocatable; least-loaded node; no solver timeout: exhaustive search (6 D3 candidates, all displacement sets) measured against 30 s / 2 s / 5 s / 60 s and flagged if exceeded (measured: ms)	Small search space -> exact answer, deadlines checked every run	vessel_baseline.yaml:145, placement.py::find_feasible_placement, admission.py:1000, arbitration.py:646,796, degradation.py:374-375
Mission priorities / crew	Weights per mission; confirmation delays; actions forbidden at 0	survey_primary / rescue_primary weights and shed order; D3 utility = survey continuity + 2.2 x landing margin; confirm 30 s (14) / 120 s (6) / automated (0); at 0 crew: no deck clearing, no physical install	Swappable file decides; safety constraints are never weighed	data/priorities/*.yaml, arbitration.py:447, scheduling.py:25-30, admission.py:358
Anything else	Guest behaviour beyond the card	G4 draws 0 W from the ship while flying, all via hangar station; airborne G4 / submerged G3 never shed by D4; G5 power split container/antenna; G6 card (420 kg) invented	Cards silent on these	ASSUMPTIONS.yaml, degradation.py:14 DEPLOYED_VEHICLE_MODES, G6_*.yaml

Scenario B ledger (G1+G3+G4+G5, actual / committed)

Power 57.5 / 81.5 kW -> 82.5 / 106.5 kW at T=480 s (G4 35 kW reserved) -> genset trip at T=720 s: 106.5 > 100 kW -> G5 45 -> 8 kW -> 45.5 / 69.5 kW. Water 103.8 -> 67.5 kW committed (<= 72 kW even with loop loss at 30 C). Air: mast_3 1,500 W (G1) + roof 1,200 -> 500 W (G5). Only two 4-guest sets pass mass; {G1,G2,G3,G4} = 10,035 kg fails.

What would change if the vessel had one single platform modification (mast_4)?

Closing question: we would add one more guest mast slot, mast_4 (+1 slot: <= 60 kg, <= 0.6 m2, +2,000 W forced air). Today G1 (1,500 W) and the G5 antenna (1,200 W) cannot share mast_3 (2,700 W > 2,000 W), so the antenna goes on the hangar roof and every G4 recovery depends on a roof-clearance condition. With mast_4 the antenna moves off the roof (9,853 -> 9,898 kg-eq, still 2,102 kg-eq margin), the drone conflict disappears and D1/D3 lose a conditional rule. It does not fix mass: {G1,G2,G3,G4} (10,035 kg), {G1,G2,G3,G5} and {G2,G3,G4,G5} still fail. The same gain applies to {G1,G2,G4,G5} (7,853 -> 7,898 kg-eq).